Lecture 39 — Least squares, data fitting, and positive definiteness

Wed Dec 2

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Boyd video (optional): Lecture 34 — least squares examples · Lecture 35 — least-squares data fitting · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, and Lachniet Ch. 2 (least-squares / data-fitting exercises). Related objectives: Lectures. Schedule.

Bring paper and a laptop if you have one. Warm-up, fill-in, a small real dataset on the board/software, then **Your turn**. Full least-squares software practice is on Friday (see the schedule).

Learning objectives

By the end of class, you should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of A.
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. Recognize that choosing different basis columns for A (monomials vs a truncated-power / knot basis) changes the model, not the LS geometry.
7. Recognize that $A^\mathsf{T}A$ is symmetric positive semidefinite, explain $x^\mathsf{T}A^\mathsf{T}Ax = \|Ax\|^2$, and recognize that $A^\mathsf{T}A$ is positive definite when A has linearly independent columns.

Warm-up

Monday: $\hat{b} = A\hat{x} = \text{proj}_{C(A)}(b)$, so the residual $r = b - \hat{b}$ is orthogonal to every column of A in the Lecture 4 sense: each $a_j^\mathsf{T} r = 0$. Stacked, that is $A^\mathsf{T} r = 0$, which rearranges to the **normal equations**

$$A^\mathsf{T} A\,\hat{x} = A^\mathsf{T} b.$$

(a) If A is $m \times n$ with $m > n$, what size is $A^\mathsf{T}A$? Why is that nicer than solving $Ax = b$ directly?

(b) True or false: $A^\mathsf{T}A$ is always symmetric.

Fill in

From projection to algebra (no calculus). "$r \perp C(A)$" means $A^{\mathsf{T}} r = 0$. Substitute $r = b - A\hat{x}$ to get $A^{\mathsf{T}} A\hat{x} = A^{\mathsf{T}} b$. So the normal equations are the algebraic form of the projection picture — we never need derivatives of $\|Ax - b\|^2$; the Lecture 4 right-angle condition is enough.

Normal equations / software. Solving $A^{\mathsf{T}} A\hat{x} = A^{\mathsf{T}} b$ (when A has independent columns) gives the unique least-squares $\hat{x}$. In practice we often call a solver instead of forming $A^{\mathsf{T}} A$ by hand:

Task	Octave	Python
Least-squares $\hat{x}$	`xhat = A \ b`	`xhat = np.linalg.lstsq(A, b, rcond=None)[0]`
Fitted values	`bhat = A*xhat`	`bhat = A @ xhat`
Residual	`r = b - bhat`	`r = b - bhat`
RMS	`rms = norm(r)/sqrt(m)`	`rms = np.linalg.norm(r)/np.sqrt(m)`

Orthogonality check. At a least-squares solution, $A^{\mathsf{T}} r \approx$ _____ (numerical noise ok).

RMS. Root-mean-square residual RMS $= \|r\|/\sqrt{m}$ reports a typical per-equation error in the same units as b.

Useful vs coefficients. Always ask: do fitted values track the data? Is RMS small *in context*? Blindly quoting $\hat{x}$ is not enough.

Basis choice = columns of A (Lecture 14 vocabulary). A "nice" curve is often approximated by a **finite** combination of basis functions. Two useful choices:

Basis idea	Typical columns of A (one row per sample t_i)	Good for
Monomials	1, t, t^2, …	Smooth global trends (polynomials)
Truncated powers (order 1)	1, t, $(t-\xi)_+$ where $(t-\xi)_+ = \max(t-\xi, 0)$	A **kink / change of slope** at a knot ξ

Same LS projection onto $C(A)$ either way — you just changed the subspace by changing the basis. Higher-order truncated powers $(t-\xi)_+^k$ exist; we only need the picture that "turn on after ξ" can beat a global polynomial when the data bend once.

Neural networks (one honest sentence). A **ReLU** unit is the truncated-linear ray $(z)_+ = \max(z, 0)$. With a **bias**, $\mathrm{ReLU}(t-\xi) = \max(t-\xi, 0)$ is exactly our order-1 truncated power with knot ξ — the bias **chooses where the ray starts**. In a network those knots / directions are **data-driven** (learned), then combined with coefficients, just as we combine columns of A in least squares. Stacking enough such ReLU features is a standard route to **universal approximation**: rich enough ramp bases can approximate a wide class of nice functions. We will not prove that theorem; the takeaway is that "choose a basis, fit coefficients" is the same skeleton — monomials, fixed knots, or learned ReLU biases.

Where this shows up in the news. Large language models such as ChatGPT are built from **neural networks**. The "**T**" in GPT stands for **Transformer** — a particular network architecture (attention layers stacked deep), not a different kind of mathematics. Same broad idea: learned features + linear combinations at enormous scale. We will not study Transformers in this course.

Positive semidefinite / definite (Boyd / class notes). A symmetric matrix P is **positive semidefinite (PSD)** if $x^{\mathsf{T}} P x \geq 0$ for all x; **positive definite (PD)** if $x^{\mathsf{T}} P x > 0$ for all $x \neq 0$.

For any A:

$$x^{\mathsf{T}}(A^{\mathsf{T}} A)x = \|Ax\|^2 \geq 0,$$

so $A^\mathsf{T}A$ is always _____. It is **positive definite** precisely when $Ax = 0 \Rightarrow x = 0$, i.e. when A has _____ columns — then the normal equations have a unique solution.

Worked in class

W1. Tiny fit by hand / software. Four campus shuttle wait times (minutes) at hours $t = 1, 2, 3, 4$ after 8am:

$$b = (6,\ 7,\ 10,\ 11).$$

Model a line $b \approx x_1 + x_2 t$ with

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}.$$

(a) Form $A^\mathsf{T}A$ and $A^\mathsf{T}b$ (or use `A\b` / `lstsq`).
(b) Report $\hat{x} = (x_1, x_2)$ and the fitted values $\hat{b} = A\hat{x}$.
(c) Compute r and RMS. In one sentence: is a linear trend reasonable here?

Expected (for checking): $\hat{x} = (4,\ 1.8)$ exactly from the normal equations; $\hat{b} = (5.8,\ 7.6,\ 9.4,\ 11.2)$; RMS $= \sqrt{0.2} \approx 0.45$ minutes.

W2. Orthogonality. With your $\hat{x}$ from W1, compute $A^\mathsf{T}r$. Should it be near $(0, 0)$?

W3. Plot story. On axes "hour t" vs "wait (min)", sketch data points and the fitted line. Where is the residual for $t = 1$ visible?

W4. PSD identity. Expand $x^\mathsf{T} A^\mathsf{T} A x$ for $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$ and a general $x = (x_1, x_2)$. Confirm it equals $\|Ax\|^2$ and is ≥ 0.

W5. PD vs PSD. Explain why $A^\mathsf{T} A$ fails to be positive definite if the two columns of A are equal. What goes wrong for uniqueness of $\widehat{x}$?

W6. Truncated power vs monomials. Five samples $t = 1, 2, 3, 4, 5$ with responses

$$b = (2,\ 2.2,\ 2.4,\ 5,\ 7)$$

(almost flat, then steeper after $t = 3$).

(a) Build the **line** model: columns 1 and t. (Optional: glance at RMS after `lstsq`.)

(b) Build a **truncated-power** model with knot $\xi = 3$:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 3 & 0 \\ 1 & 4 & 1 \\ 1 & 5 & 2 \end{bmatrix}$$

(columns 1, t, $(t-3)_+$). Solve for $\widehat{x}$ and compare RMS to the line. Which subspace matches the kink better?

Your turn

Y1. Using the W1 data, suppose software returns $\hat{x} = (4,\ 1.8)$. Compute $\hat{b}_1 = x_1 + x_2 \cdot 1$ and the first residual $r_1 = b_1 - \hat{b}_1$.

Y2. True or false: $A^{\mathsf{T}}A$ is always positive definite for every A.

Y3. Why check $A^{\mathsf{T}}r \approx 0$ after a least-squares solve?

Y4. A fit has RMS = 0.02 °C for outdoor temperature data that typically varies by 10 °C across the day. Useful model, or not? One sentence.

Y5. In one sentence: what does switching from monomials to a truncated-power column change — the LS geometry, or the model subspace $C(A)$?

Answers (Your turn)

Y1. $\hat{b}_1 = 4 + 1.8 = 5.8$; $r_1 = 6 - 5.8 = 0.2$.

Y2. **False.** $A^\mathsf{T}A$ is always PSD, but PD only when A has linearly independent columns (equivalently $\ker A = \{0\}$).

Y3. It is the normal-equation / geometric check that the residual is orthogonal to $C(A)$ — evidence that $\hat{x}$ really is a least-squares critical point (up to roundoff).

Y4. **Useful** in context: RMS $\ll$ the natural scale of the data, so typical pointwise error is tiny compared with day-scale variation.

Y5. The LS geometry is the same (project b onto $C(A)$); what changes is the **choice of subspace** — different basis columns mean a different $C(A)$.